



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Subject: Programming with Java

ASSIGNMENT No :2

TITLE :Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Assignment 2: Methods and Static Members

Problem Statement :

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Problem Code:

```

J EXP_2.java > EXP_2 > main(String[])
1  class Employee {
2
3      // INSTANCE VARIABLES
4      int empid;
5      String empname;
6      double salary;
7
8      // STATIC VARIABLE
9      static String company = "Google";
10
11     // Default Constructor
12     Employee() {
13         // Java automatically assigns:
14         // empid = 0
15         // empname = null
16         // salary = 0.0
17     }
18
19     // Parameterized Constructor
20     Employee(int empid, String empname, double salary) {
21         this.empid = empid;
22         this.empname = empname;
23         this.salary = salary;
24     }
25
26     // Copy Constructor
27     Employee(Employee e) {
28         this.empid = e.empid;
29         this.empname = e.empname;
30         this.salary = e.salary;
31     }
32
33     // Instance Method
34     void displayDetails() {
35         System.out.println("Employee ID:          " + empid);
36         System.out.println("Employee Name:          " + empname);
37         System.out.println("Employee Salary:       " + salary);
38         System.out.println("Company:                " + company);
39         System.out.println(x: "*****");
40         System.out.println(x: "*****");
41     }
42
43     // Instance Method
44     void increaseSalary(double amount) {
45         salary += amount;

```

```
J EXP_2.java > EXP_2 > main(String[])
1  class Employee {
44      void increaseSalary(double amount) {
45          salary += amount;
46      }
47
48      // Static Method
49      static void showCompany() {
50          System.out.println("Company Name: " + company);
51          System.out.println();
52      }
53  }
54
55  public class EXP_2 {
56
57      Run | Debug
58      public static void main(String[] args) {
59          Employee.showCompany();
60          // Default Constructor
61          Employee e1 = new Employee();
62
63          // Parameterized Constructor
64          Employee e2 = new Employee(empid: 101, empname: "Rahul", salary: 55000);
65
66          // Copy Constructor
67          Employee e3 = new Employee(e2);
68
69          // Increase Salary
70          e2.increaseSalary(amount: 5000);
71
72          System.out.println(x: "Employee 1 (Default Constructor)");
73          e1.displayDetails();
74
75          System.out.println(x: "Employee 2 (Parameterized Constructor)");
76          e2.displayDetails();
77
78          System.out.println(x: "Employee 3 (Copy Constructor)");
79          e3.displayDetails();
80      }
81  }
```

OUTPUT:

```

PS C:\Users\kchig\OneDrive\Desktop\java_college> cd "c:\Users\kchig\OneD
Company Name: Google

Employee 1 (Default Constructor)
Employee ID:      0
Employee Name:    null
Employee Salary:  0.0
Company:         Google
*****
*****
Employee 2 (Parameterized Constructor)
Employee ID:      101
Employee Name:    Rahul
Employee Salary:  60000.0
Company:         Google
*****
*****
Employee 3 (Copy Constructor)
Employee ID:      101
Employee Name:    Rahul
Employee Salary:  55000.0
Company:         Google
*****
*****
PS C:\Users\kchig\OneDrive\Desktop\java_college>

```

Learning Objectives :

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory :

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Outcome:

Successfully developed a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization. The program also implemented instance methods and static members for managing object data, creating duplicate objects, and performing basic operations.

Conclusion :

The assignment helped in understanding the role of constructors in object creation and initialization. It demonstrated the use of default constructors for automatic initialization, parameterized constructors for user-defined initialization, and copy constructors for creating duplicate objects. The implementation also reinforced the concepts of methods, static variables, and static methods, thereby strengthening the understanding of object-oriented programming principles in Java.

Exercise :

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

Program Code:

```
J EXP_2_1.java > EXP_2_1 > main(String[])
1  class Student {
2      int rollNo;
3      String name;
4      double marks;
5
6      static String college = "SIT Pune";
7
8      Student() {
9
10     }
11
12     Student(int rollNo, String name, double marks) {
13         this.rollNo = rollNo;
14         this.name = name;
15         this.marks = marks;
16     }
17
18     Student(Student s) {
19         this.rollNo = s.rollNo;
20         this.name = s.name;
21         this.marks = s.marks;
22     }
23
24     void displaydetails() {
25         System.out.println("Student College: " + college);
26         System.out.println("Student Roll Number: " + rollNo);
27         System.out.println("Student Name: " + name);
28         System.out.println("Student marks: " + marks);
29         System.out.println(x: "*****");
30         System.out.println(x: "*****");
31     }
32
33     void updatemarks(double bonus) {
34         marks = marks + bonus;
35     }
36
37     static void collegename() {
38         System.out.println("College is: " + college);
39         System.out.println();
40     }
41 }
```

```

42
43 public class EXP_2_1 {
    Run | Debug
44     public static void main(String[] args) {
45         Student.collegename();
46
47         Student s1 = new Student();
48
49         Student s2 = new Student(rollNo: 68, name: "Gaurav Singh", marks: 85.5);
50
51         s2.updatemarks(bonus: 4.5);
52
53         Student s3 = new Student(s2);
54
55         System.out.println(x: "-----Student s1 Details-----");
56         s1.displaydetails();
57
58         System.out.println(x: "-----Student s2 Details-----");
59         s2.displaydetails();
60
61         System.out.println(x: "-----Student s3 Details-----");
62         s3.displaydetails();
63
64     }
65 }

```

Output:

```

PS C:\Users\kchig\OneDrive\Desktop\java_college> cd
College is: SIT Pune

-----Student s1 Details-----
Student College:      SIT Pune
Student Roll Number:  0
Student Name:         null
Student marks:        0.0
*****
*****
-----Student s2 Details-----
Student College:      SIT Pune
Student Roll Number:  68
Student Name:         Gaurav Singh
Student marks:        90.0
*****
*****
-----Student s3 Details-----
Student College:      SIT Pune
Student Roll Number:  68
Student Name:         Gaurav Singh
Student marks:        90.0
*****
*****
PS C:\Users\kchig\OneDrive\Desktop\java_college>

```

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

Program Code:

J EXP_2_2.java > EXP_2_2 > main(String[])

```
1  class MobilePhone {
2      int phoneId;
3      String brand;
4      String model;
5      double price;
6
7      static String company = "Samsung";
8
9      MobilePhone() {
10
11     }
12
13     MobilePhone(int phoneId, String brand, String model, double price) {
14         this.phoneId = phoneId;
15         this.brand = brand;
16         this.model = model;
17         this.price = price;
18     }
19
20     MobilePhone(MobilePhone p) {
21         this.phoneId = p.phoneId;
22         this.brand = p.brand;
23         this.model = p.model;
24         this.price = p.price;
25     }
26
27     void displaydetails() {
28         System.out.println("Mobile Phone Company:      " + company);
29         System.out.println("Mobile Phone ID:      " + phoneId);
30         System.out.println("Mobile Phone Brand:      " + brand);
31         System.out.println("Mobile Phone Model:      " + model);
32         System.out.println("Mobile Phone Price:      $" + price);
33         System.out.println(x: "*****");
34         System.out.println(x: "*****");
35     }
36
37     void updatePrice(double amount) {
38         price = price + amount;
39     }
40
41     static void showCompanyName() {
42         System.out.println("Company of all Mobile Phones is: " + company);
43         System.out.println();
44     }
45 }
```

```
J EXP_2_2.java > EXP_2_2 > main(String[])
1  class MobilePhone {
44  |  }
45  }
46
47  public class EXP_2_2 {
    Run | Debug
48  public static void main(String[] args) {
49      MobilePhone.showCompanyName();
50
51      MobilePhone p1 = new MobilePhone();
52
53      MobilePhone p2 = new MobilePhone(phoneId: 101, brand: "Galaxy", model: "S25 Ultra", price: 129999);
54
55      MobilePhone p3 = new MobilePhone(p2);
56
57      p2.updatePrice(amount: 15000);
58
59      System.out.println(x: "-----MOBILE PHONE INVENTORY SYSTEM-----");
60
61      System.out.println(x: "***Mobile Phone 1 Details***");
62      p1.displaydetails();
63
64      System.out.println(x: "***Mobile Phone 2 Details***");
65      p2.displaydetails();
66
67      System.out.println(x: "***Mobile Phone 3 Details***");
68      p3.displaydetails();
69
70  }
71  }
72
```

Output:

```
PS C:\Users\kchig\OneDrive\Desktop\java_college> cd "c:
Company of all Mobile Phones is: Samsung

-----MOBILE PHONE INVENTORY SYSTEM-----
**Mobile Phone 1 Details**
Mobile Phone Company:      Samsung
Mobile Phone ID:           0
Mobile Phone Brand:        null
Mobile Phone Model:        null
Mobile Phone Price:        $0.0
*****
**Mobile Phone 2 Details**
Mobile Phone Company:      Samsung
Mobile Phone ID:           101
Mobile Phone Brand:        Galaxy
Mobile Phone Model:        S25 Ultra
Mobile Phone Price:        $144999.0
*****
**Mobile Phone 3 Details**
Mobile Phone Company:      Samsung
Mobile Phone ID:           101
Mobile Phone Brand:        Galaxy
Mobile Phone Model:        S25 Ultra
Mobile Phone Price:        $129999.0
*****
PS C:\Users\kchig\OneDrive\Desktop\java_college>
```